

Report_DEB

1. Importation des données

```
file_path <- here::here("mod/DEB_SASWI")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
)

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```

NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 = 8484)
seeds <- c(C1 = 3333, C2 = 6666, C3 = 1222)

text_priors <- '
Distrib (pAm,          TruncNormal_cv,          100,          0.2,          40, 2000);
Distrib (Fm,           TruncNormal,              0.1,           0.2,           0.01, 0.8);
Distrib (r_pAm_pM,     TruncNormal_cv,          0.8,           0.1,           0.1, 2);
Distrib (kap,          Uniform,                  0.1,           0.9);
Distrib (Ehp,          TruncNormal_cv,          100,           0.2,           40, 300);
Distrib (E_coc,        TruncNormal_cv,          30,            0.2,           1, 300);
Distrib (rOM_ClxFhorse, LogUniform,             0.00001,       0.3);
Distrib (f_Slope, TruncNormal_cv, 1, 1, 0, 10);
Distrib (dv, TruncNormal_cv, 1, 0.2, 0, 5);

# Calcul des vraisemblances
Distrib (Sigma_W, TruncNormal, 5, 1, 0.01, 100);'

text_likelihoood <- 'Likelihood (Weight,          Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihoood, NbIter)

# ./mcsim.DEBcaliSWI DEB_C1.in

```

OM_horse and OM_soil distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

	pAm.1.	Fm.1.	r_pAm_pM.1.	kap.1.	Ehp.1.	E_coc.1.
Result_mode	140.081000	0.64196300	0.100242000	0.899184000	198.68600	63.683700
2.5%	131.855000	0.47280600	0.100086000	0.882338000	167.24800	56.591400
97.5%	158.194850	0.74748795	0.115824500	0.899882000	232.86900	73.979500
Min	87.590600	0.09270370	0.100000000	0.390513000	78.97720	32.772900
Max	171.732000	0.79984600	0.841339000	0.900000000	264.77500	83.954200

```

Result_mean 144.187556 0.60543766 0.104220112 0.895480466 199.30387 65.216003
Result_sd    6.783018 0.06951958 0.008331541 0.006511257 17.05541 4.506849
           rOM_ClxHorse.1. f_Slope.1. Sigma_W.1.
Result_mode  0.0000133431 1.8609000 0.31043900
2.5%         0.0000179869 0.9905980 0.25706400
97.5%        0.0090747505 3.0518085 0.35981200
Min          0.0000111440 0.7019950 0.21386700
Max          0.0186332000 4.7352400 5.24867000
Result_mean  0.0036650055 1.7929604 0.30592108
Result_sd    0.0027609643 0.5178161 0.04682376

```

```

Number of observations, n = 277
Number of parameters, k = 8
Log-likelihood, LL = -144.6308
BIC = 334.2537

```

Potential scale reduction factors:

	Point est.	Upper C.I.
E_coc.1.	1.01	1.01
Ehp.1.	1.01	1.04
f_Slope.1.	1.01	1.04
Fm.1.	1.01	1.01
kap.1.	1.01	1.02
pAm.1.	1.01	1.02
r_pAm_pM.1.	1.00	1.00
rOM_ClxHorse.1.	1.00	1.01
Sigma_W.1.	1.00	1.01

Multivariate psrf

1.02

```

mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE]) # <- reconvertir en mcmc
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)

```

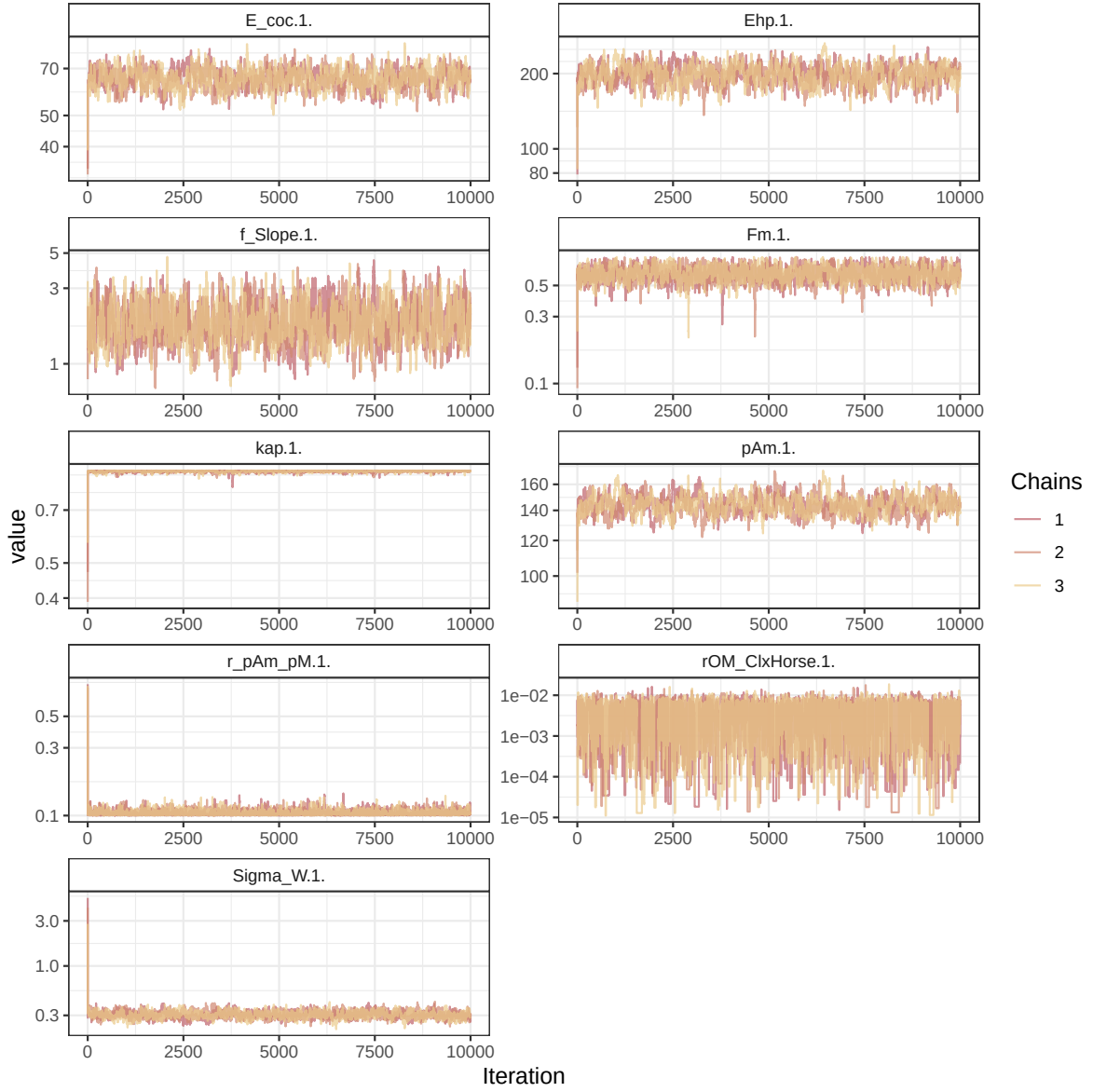


Figure 1: Parameters chains. Only the first iteration and the last `r Nb_iter_kept` are represented.

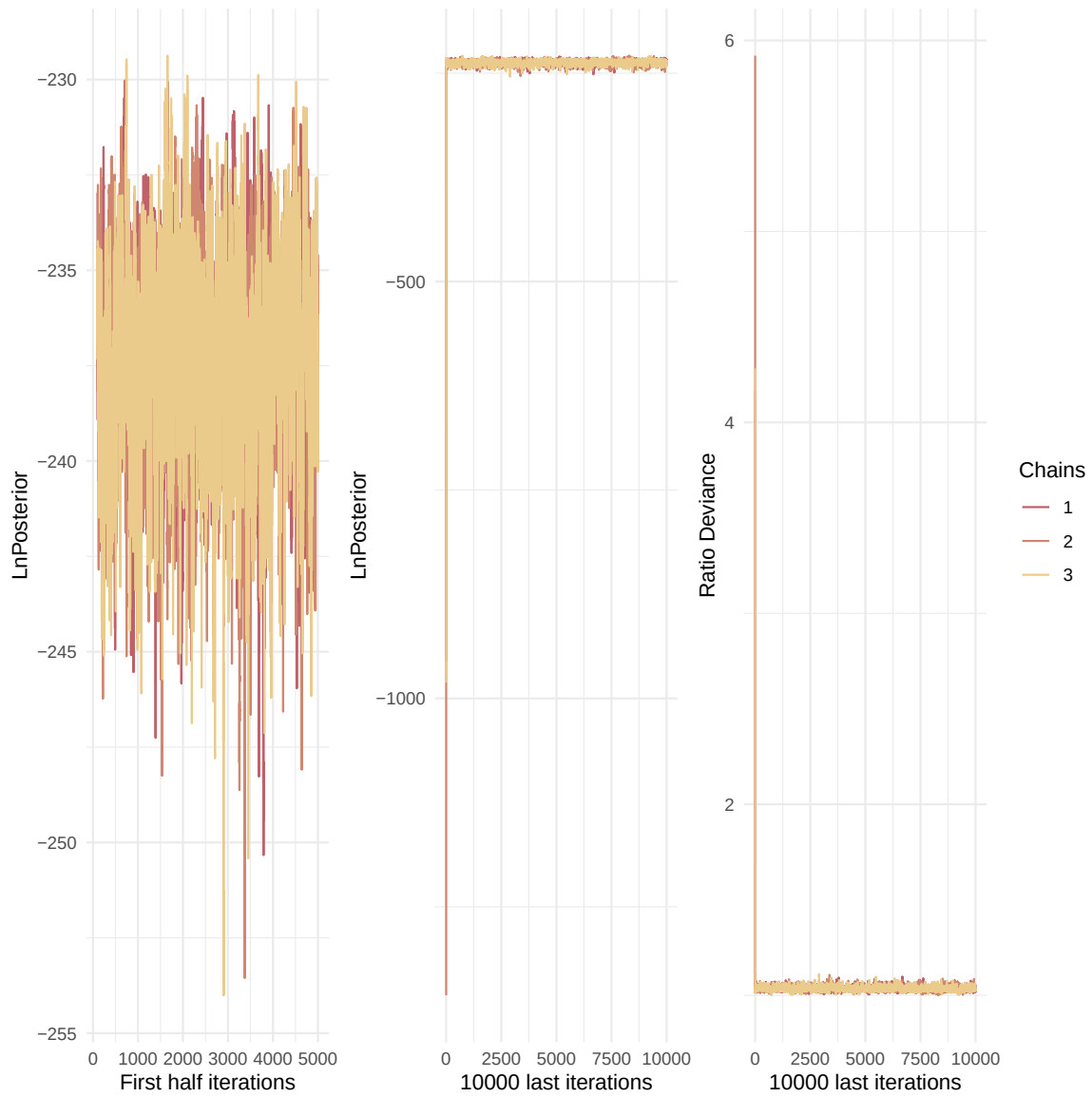


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

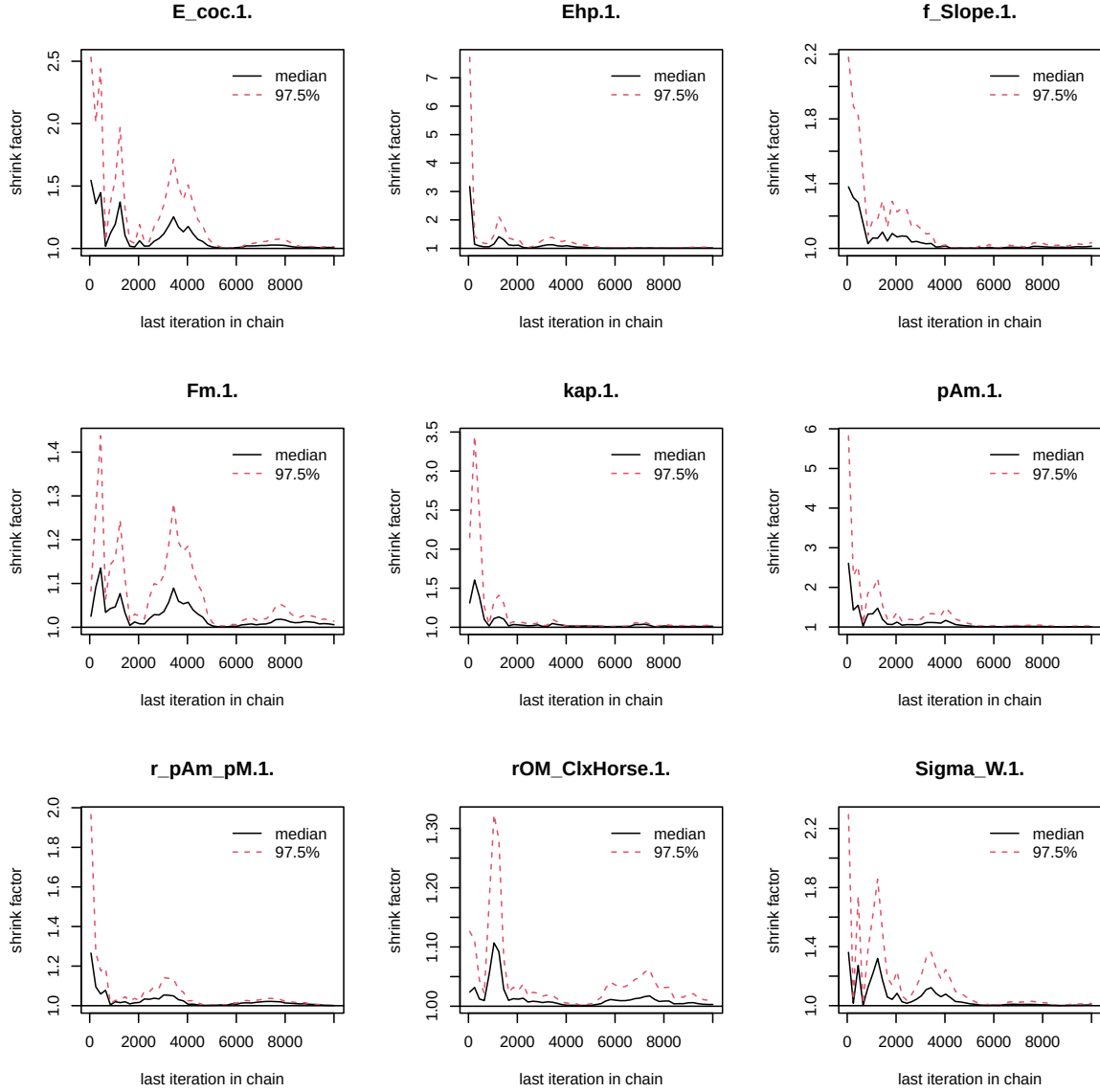


Figure 3: Chains convergence.

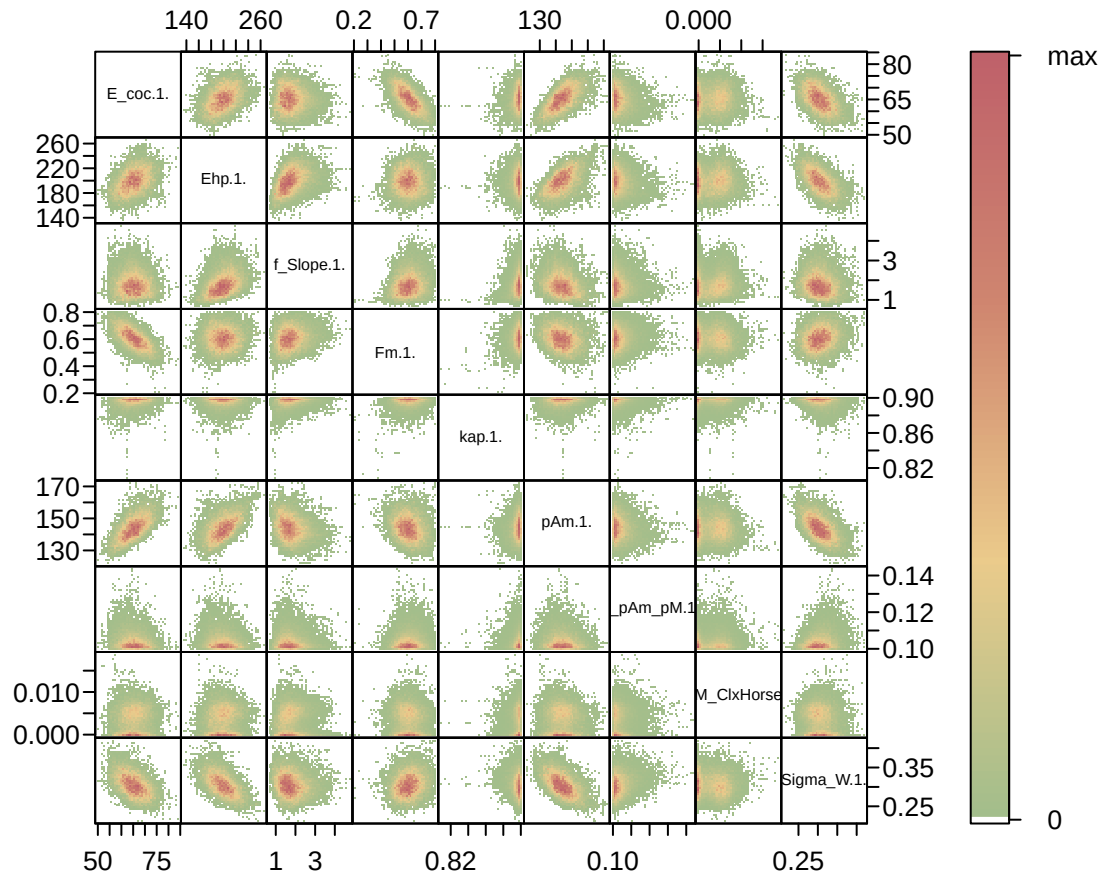


Figure 4: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
```

Posterior
and **prior** distributions

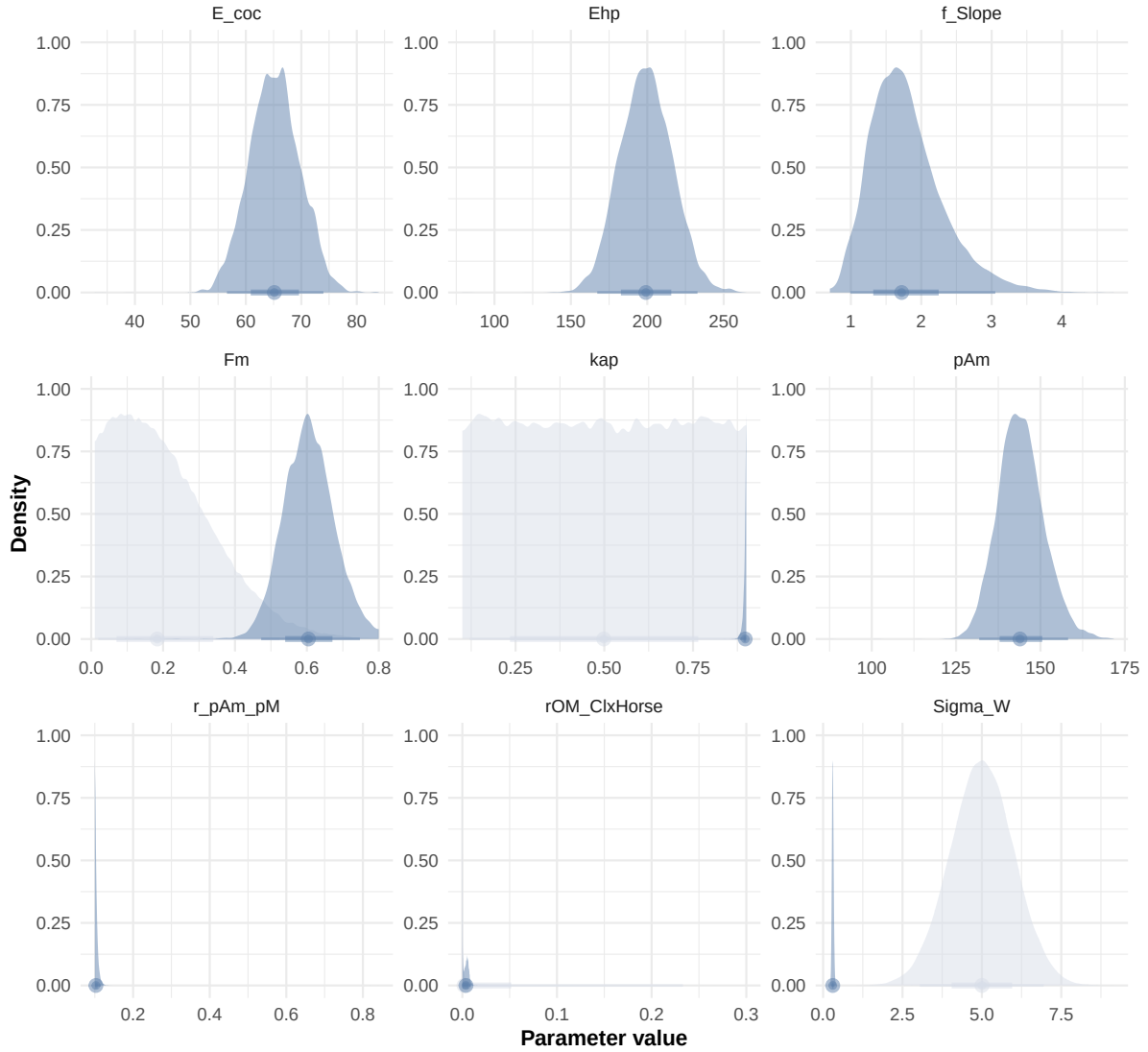


Figure 5: Distributions of the priors and the posteriors.


```

"Bart2020",
#"Gollot2026_lufa"
"Gollot2026_dens_D1",
"Gollot2026_dens_D2",
"Gollot2026_dens_D2b",
"Gollot2026_dens_D5",
"Gollot2026_dens_D7"
)

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)

text_param <- Summary_res["Result_mode", , drop = FALSE] %>%
  t() %>%
  as.data.frame() %>%
  tibble::rownames_to_column("param") %>%
  rename(val = 2) %>%
  mutate(
    param = gsub("\\\\.1\\.\\.\\.|\\.\\.\\.1\\.\\.\\.\"", "", param),
    line = glue("{param} = {val};")
  ) %>%
  pull(line) %>%
  paste(collapse = "\n")

Endpoints_print <- "Weight,
  Length_struct,
  Reproduction,
  Energy,
  Organic_matter,"

f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff, text_param, Endpoints_print)
#f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff=FALSE, text_param, text_pr

# ./mcsim.DEBcaliSWI DEB_sim_full.in

#####
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),

```

```

    color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
              Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
  )
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#####

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times)

```

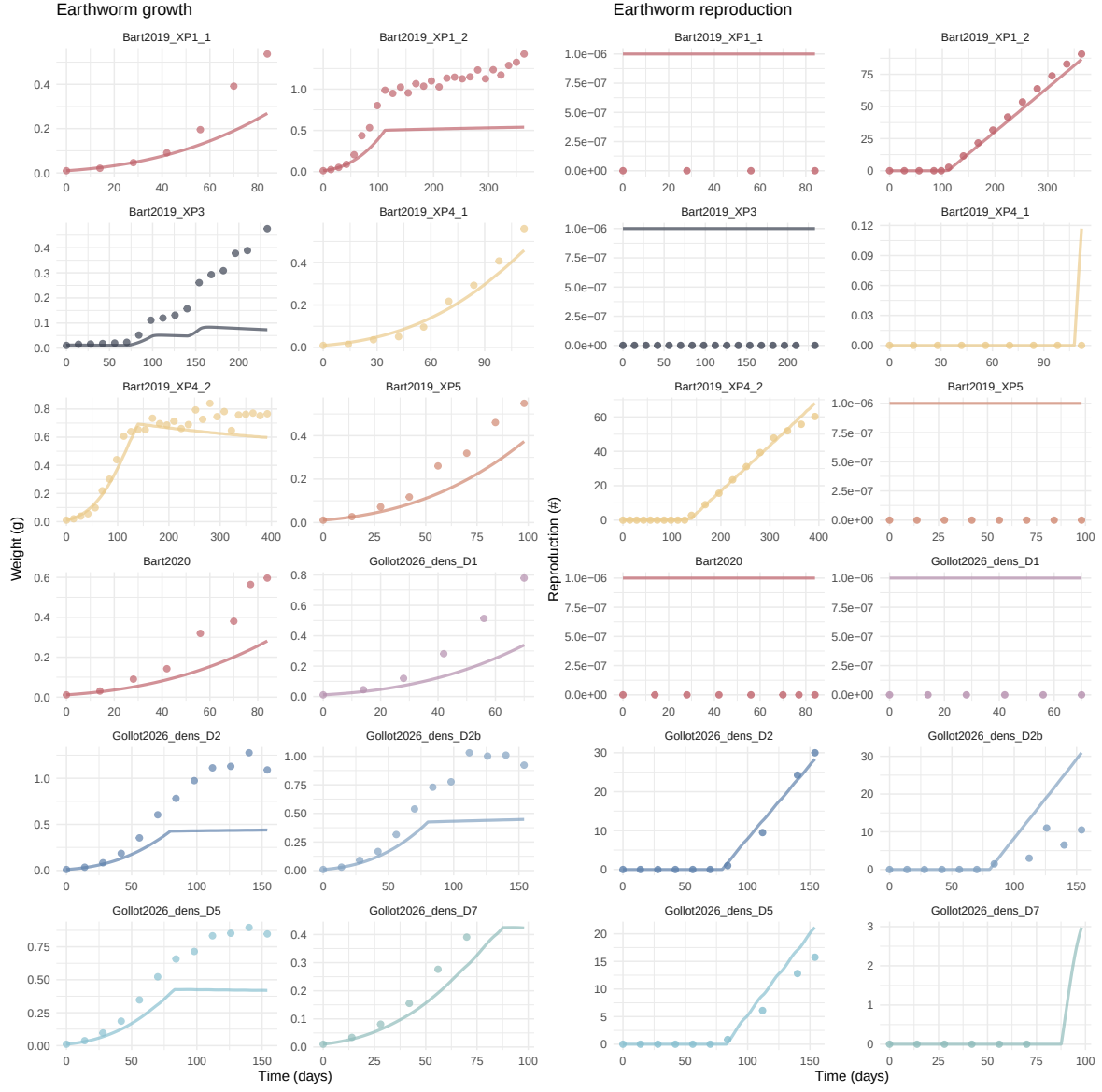
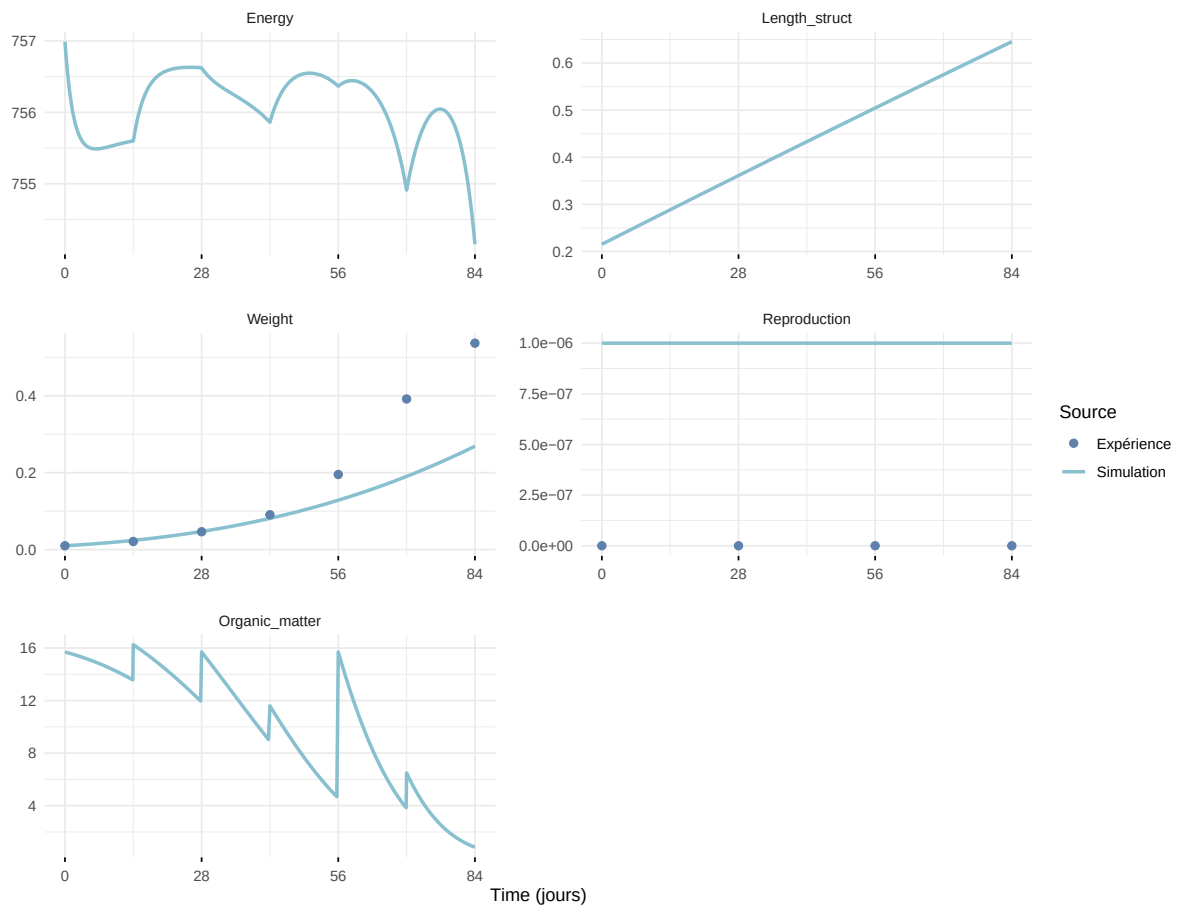
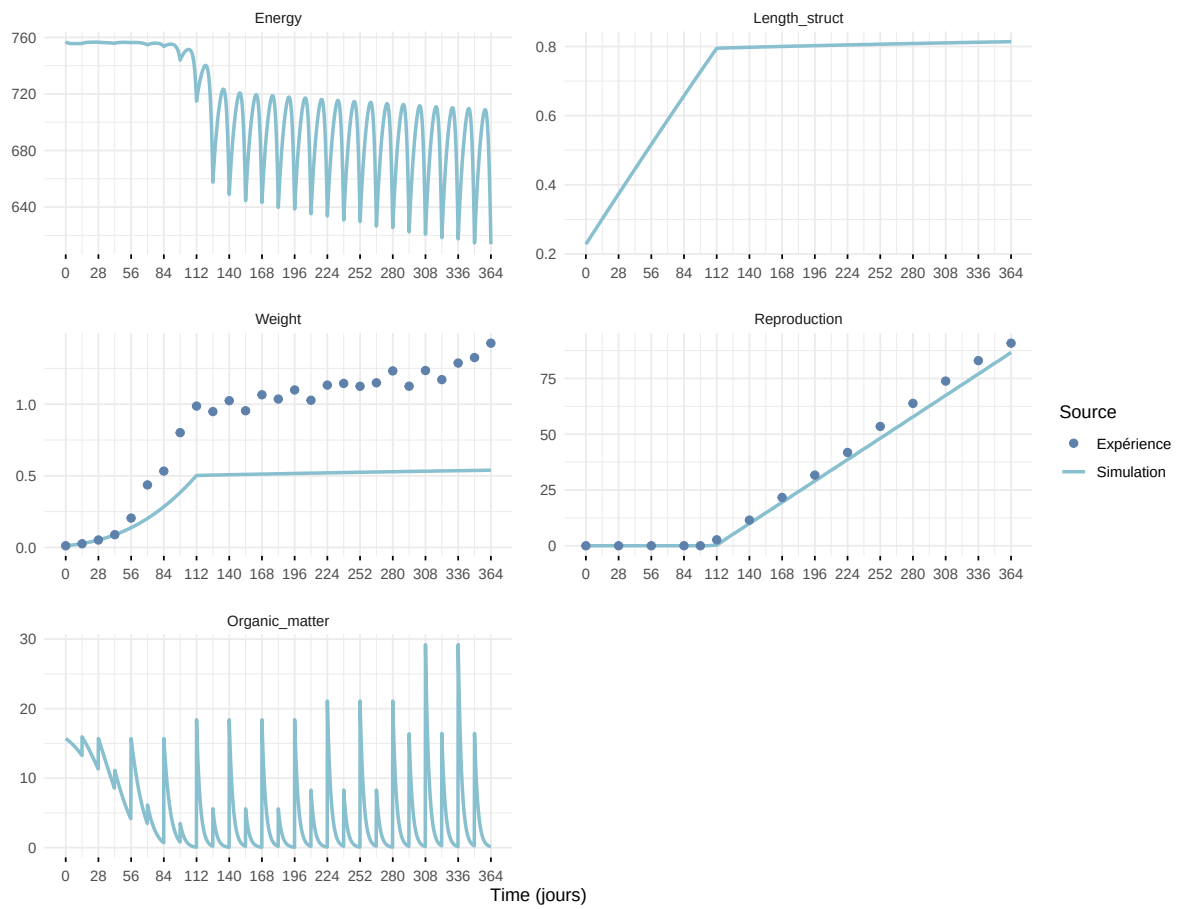


Figure 6: Simulations Weight and Reproduction.

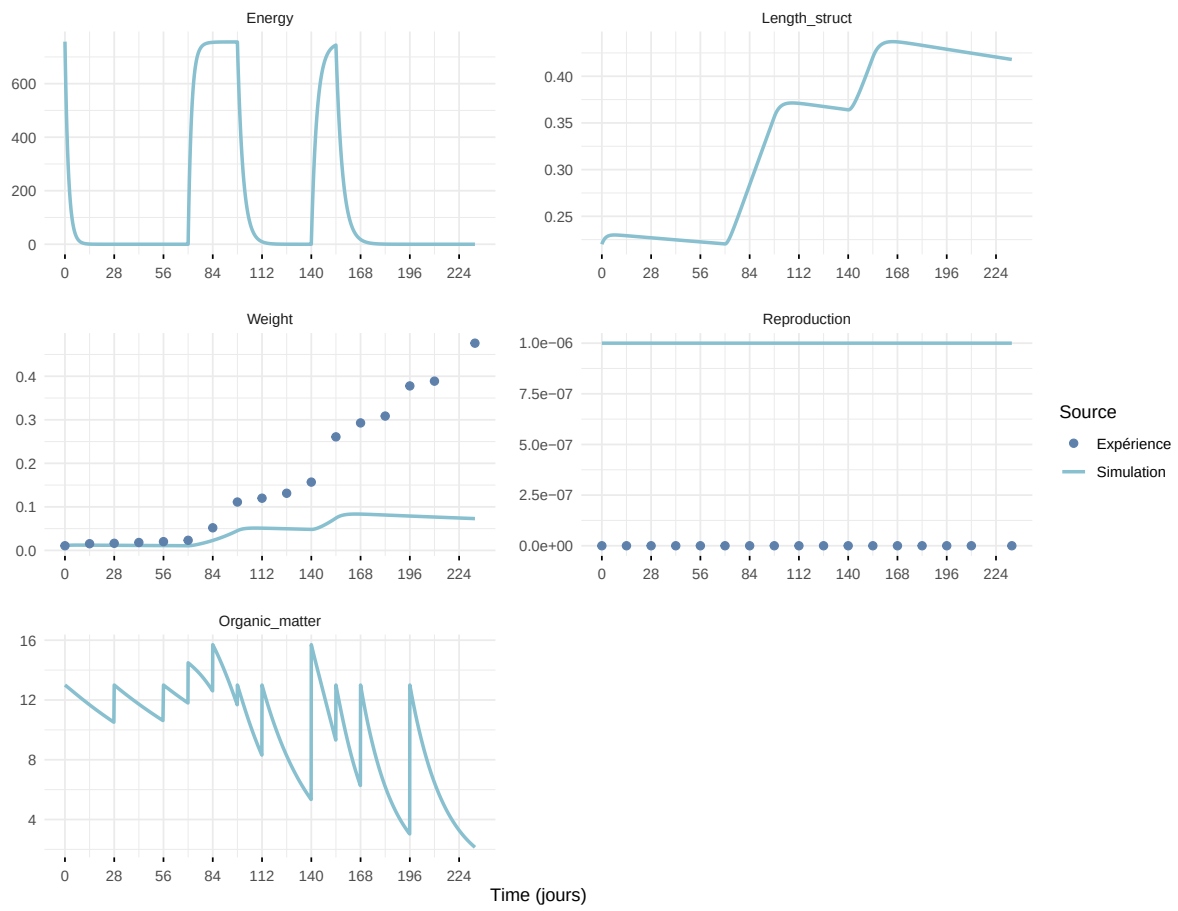
Expérience : Bart2019_XP1_1



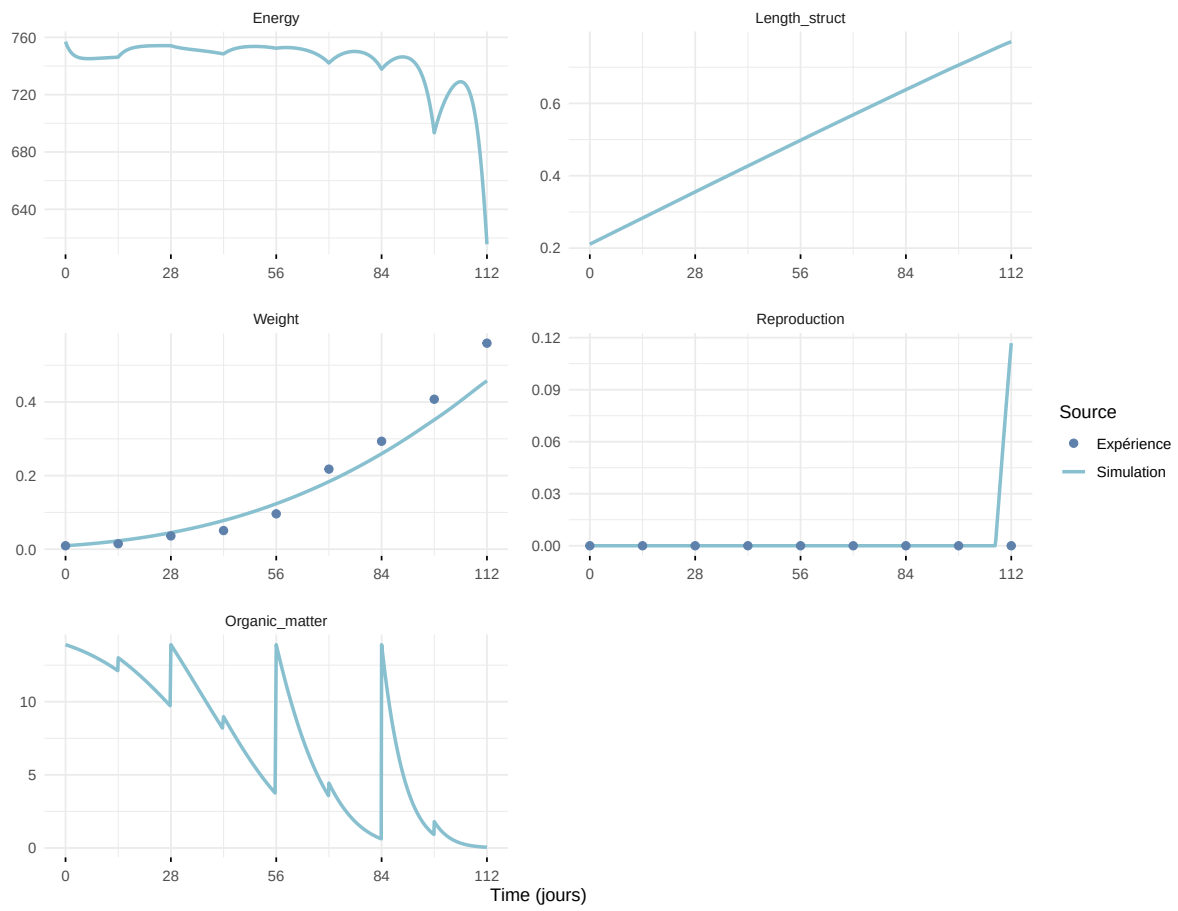
Expérience : Bart2019_XP1_2



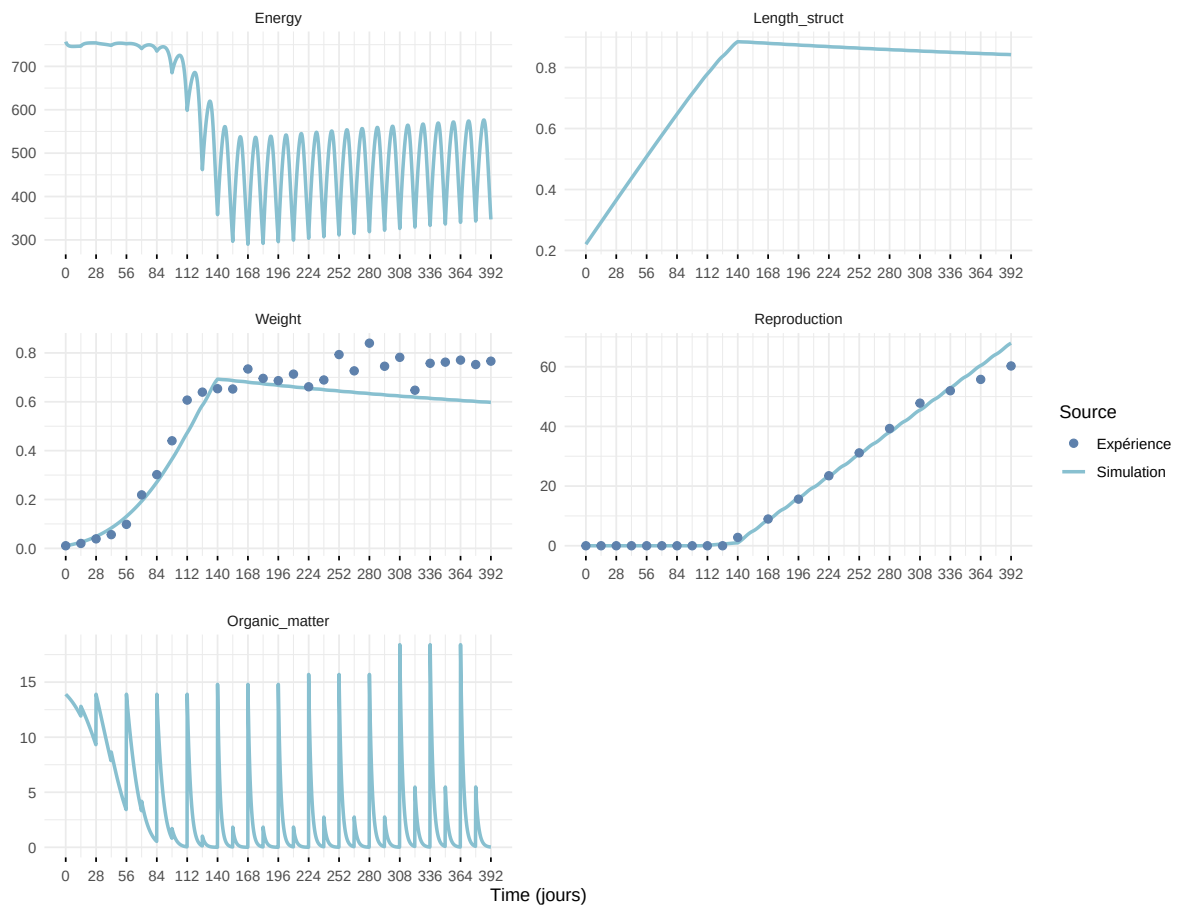
Expérience : Bart2019_XP3



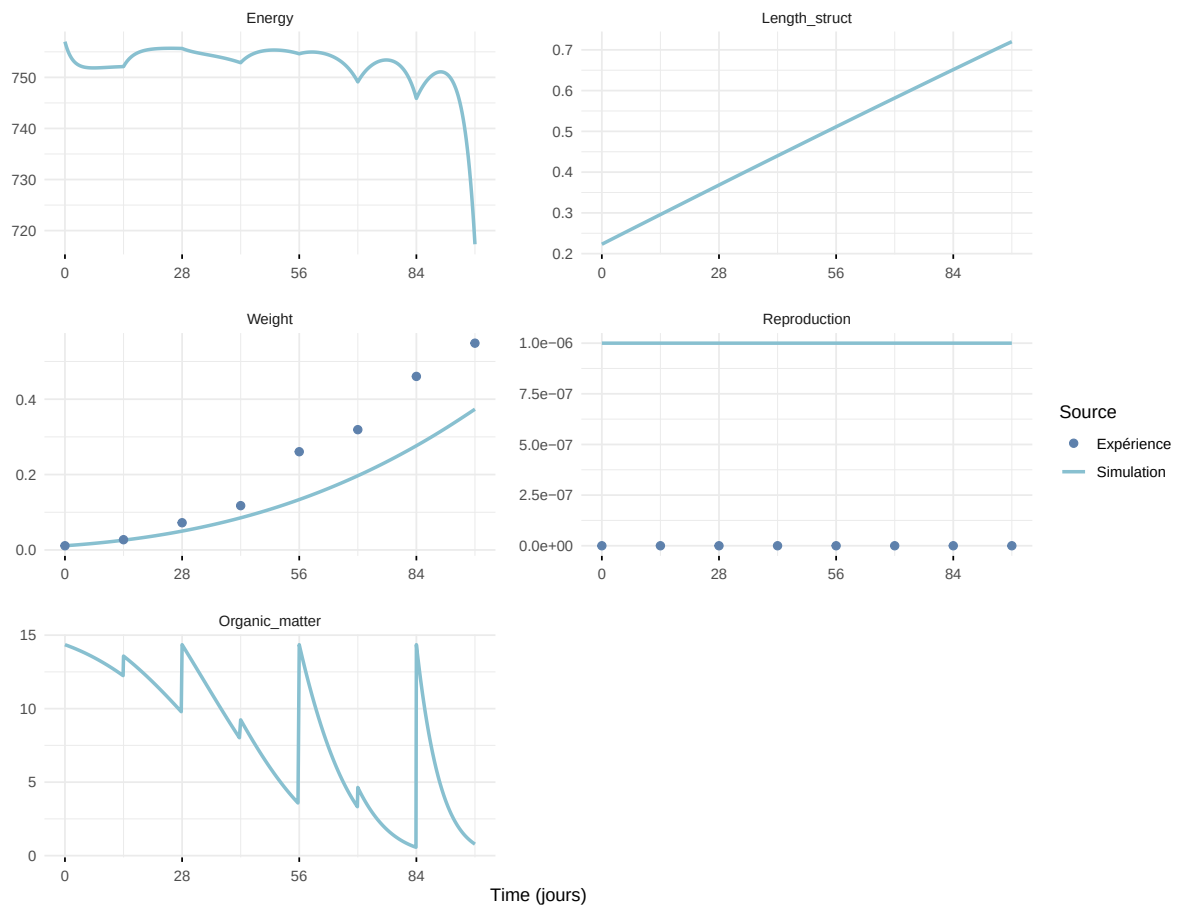
Expérience : Bart2019_XP4_1



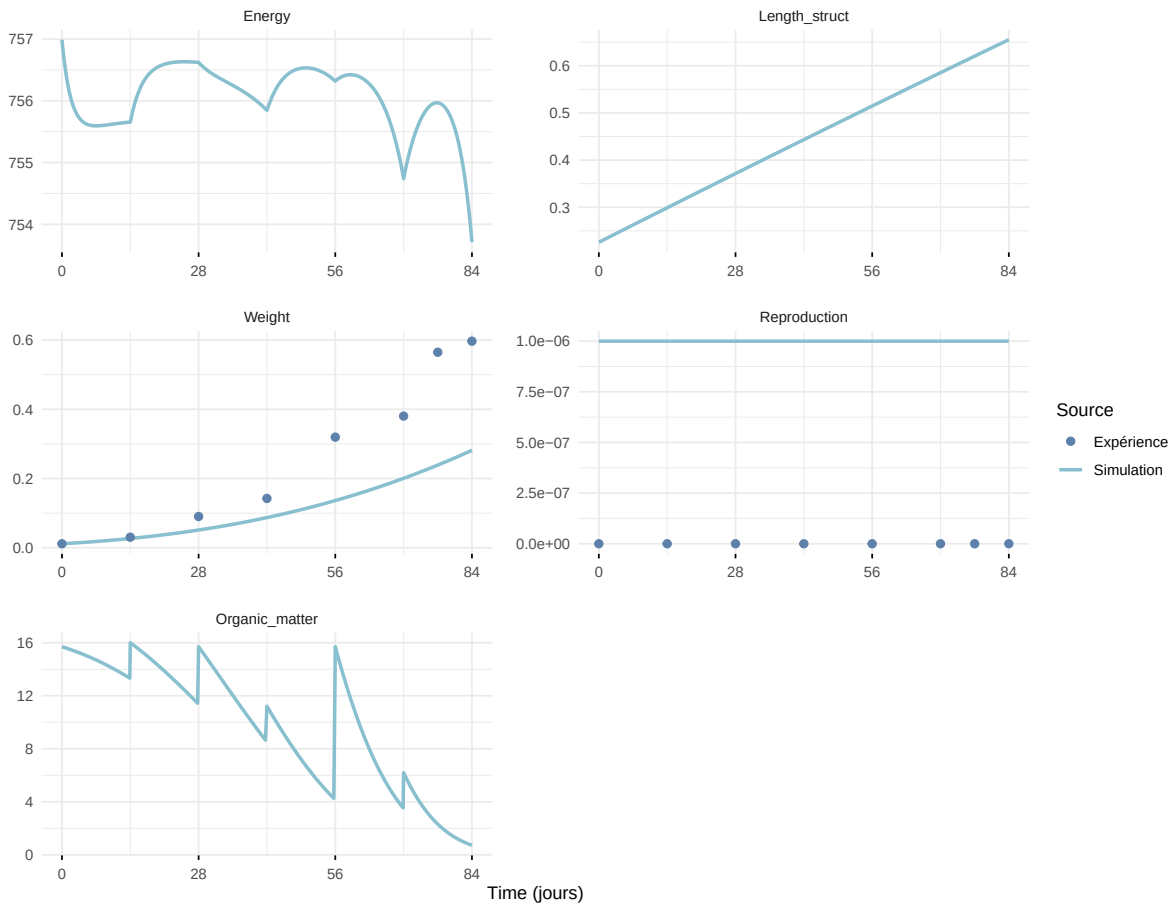
Expérience : Bart2019_XP4_2



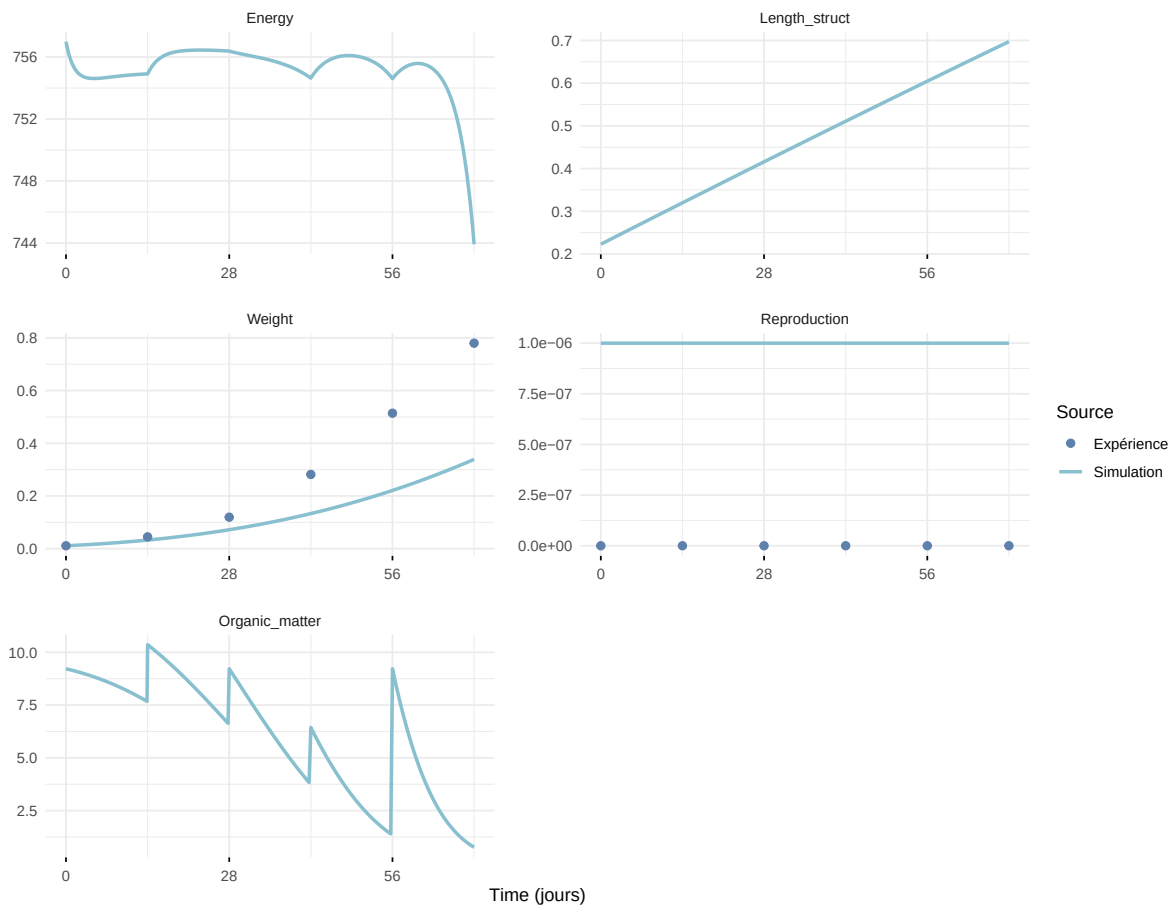
Expérience : Bart2019_XP5



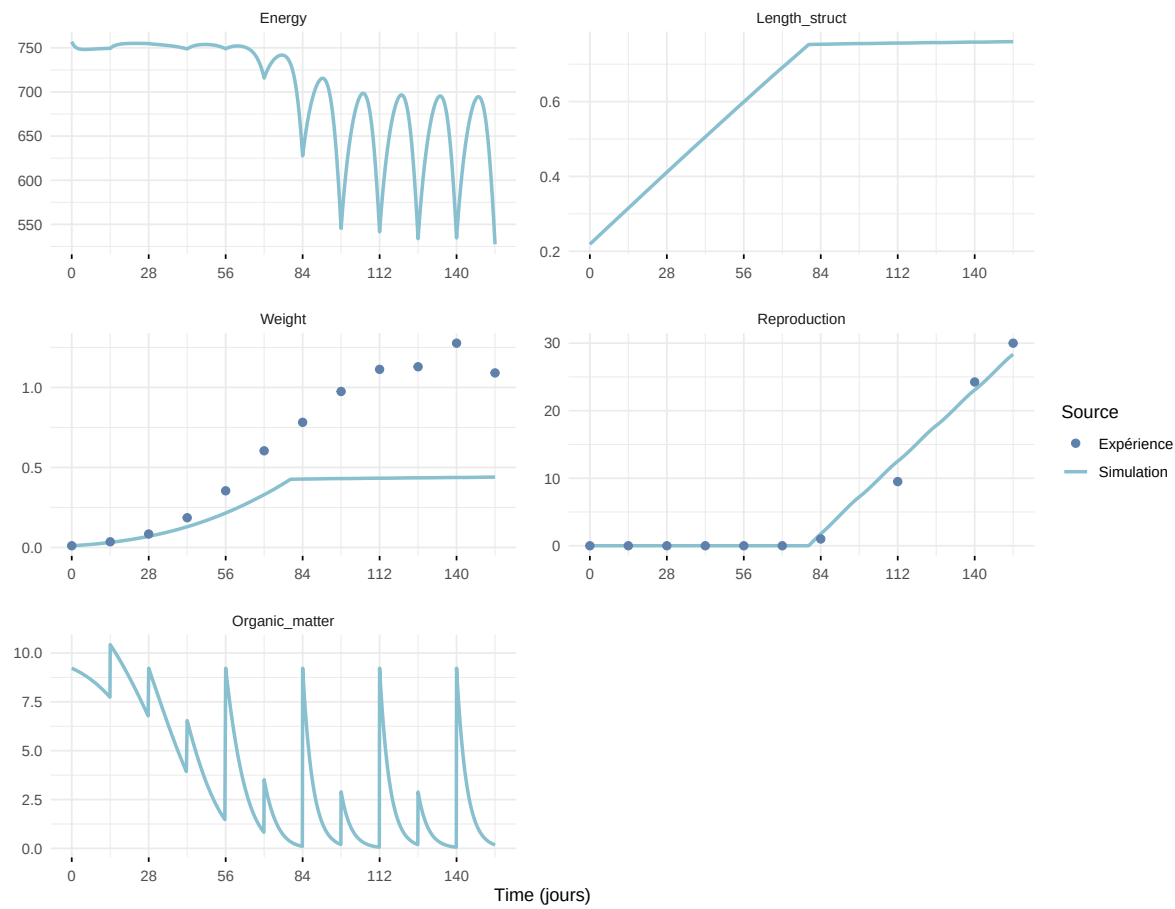
Expérience : Bart2020



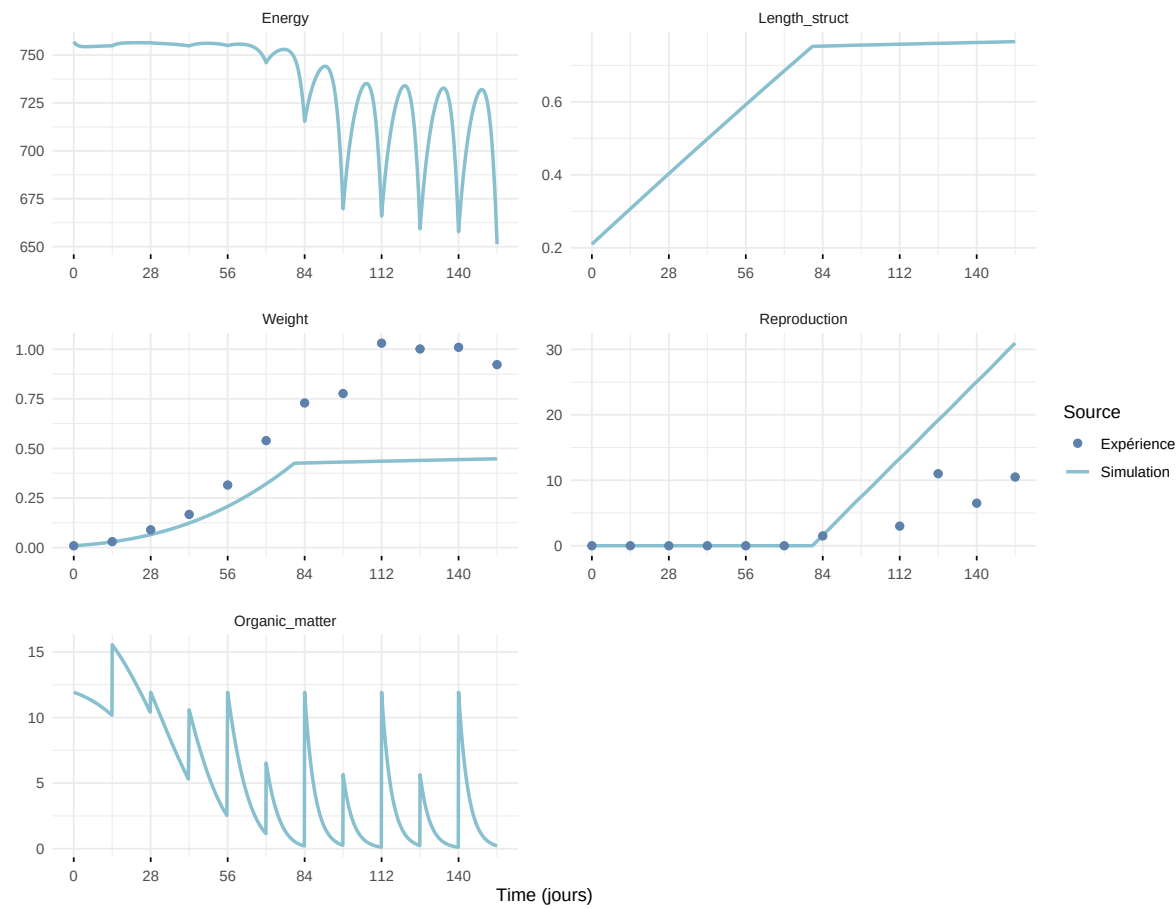
Expérience : Gollot2026_dens_D1



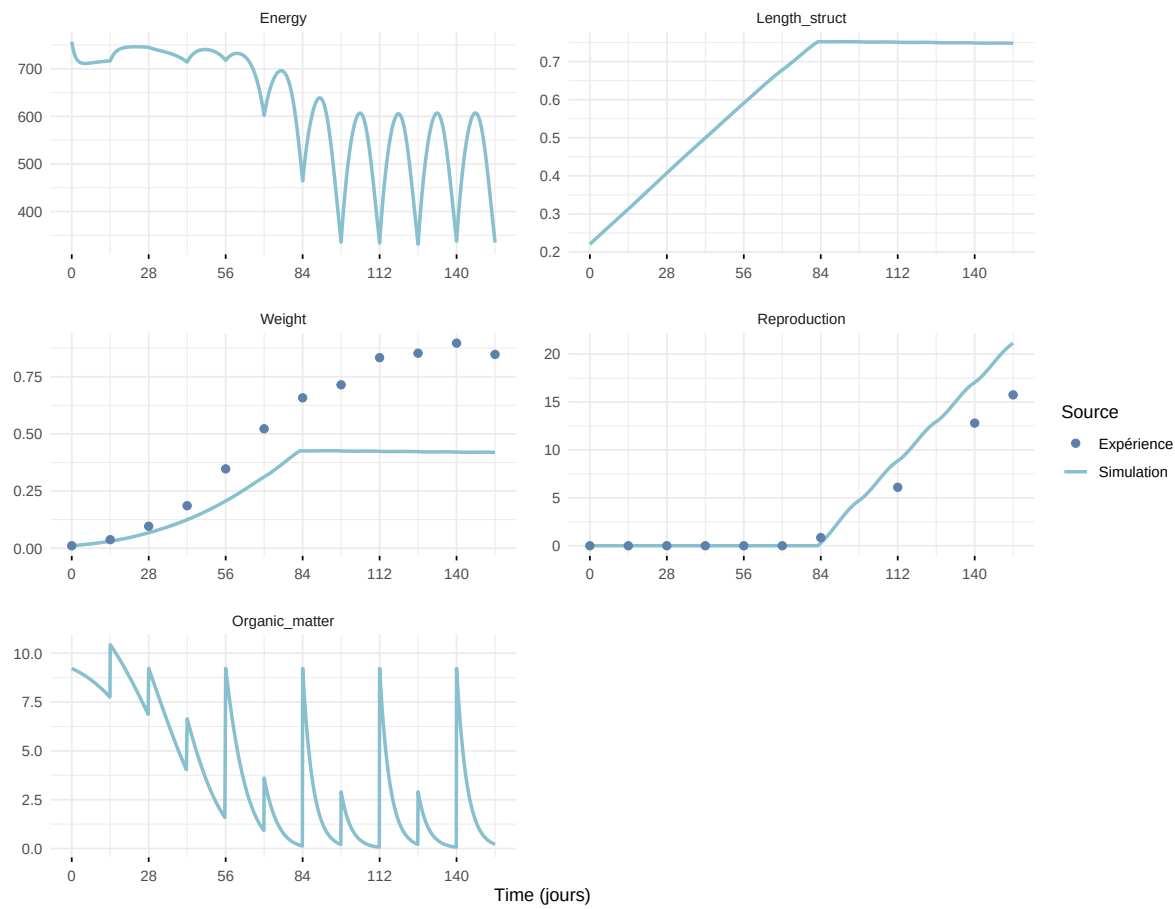
Expérience : Gollot2026_dens_D2



Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5



Expérience : Gollot2026_dens_D7

